



URL of this page: <https://medlineplus.gov/copd.html>

COPD

What is COPD?

COPD (chronic obstructive pulmonary disease) is a group of lung diseases that make it hard to breathe and get worse over time.

Normally, the airways and air sacs in your lungs are elastic or stretchy. When you breathe in, the airways bring air to the air sacs. The air sacs fill up with air, like a small balloon. When you breathe out, the air sacs deflate, and the air goes out. If you have COPD, less air flows in and out of your airways because of one or more problems:

- The airways and air sacs in your lungs become less elastic
- The walls between many of the air sacs are destroyed
- The walls of the airways become thick and inflamed
- The airways make more mucus than usual and can become clogged

What are the types of COPD?

COPD includes two main types:

- **Emphysema** [<https://medlineplus.gov/emphysema.html>] affects the air sacs in your lungs, as well as the walls between them. They become damaged and are less elastic.
- **Chronic bronchitis** [<https://medlineplus.gov/chronicbronchitis.html>] , in which the lining of your airways is constantly irritated and inflamed. This causes the lining to swell and make mucus.

Most people with COPD have both emphysema and chronic bronchitis, but how severe each type is can be different from person to person.

What causes COPD?

The cause of COPD is usually long-term exposure to irritants that damage your lungs and airways. In the United States, cigarette smoke [<https://medlineplus.gov/smoking.html>] is the main cause. Pipe, cigar, and other types of tobacco smoke can also cause COPD, especially if you inhale them.

Exposure to other inhaled irritants can contribute to COPD. These include **secondhand smoke** [<https://medlineplus.gov/secondhandsmoke.html>] , **air pollution** [<https://medlineplus.gov/airpollution.html>] , and chemical fumes or dusts from the environment or workplace.

Rarely, a genetic condition called **alpha-1 antitrypsin deficiency** [<https://medlineplus.gov/alpha1antitrypsindeficiency.html>] can play a role in causing COPD.

Who is more likely to develop COPD?

Certain factors make you more likely to develop COPD. These include:

- **Smoking.** This is the main risk factor. Up to 75% of people who have COPD smoke or used to smoke.
- **Long-term exposure to other lung irritants**, such as secondhand smoke, air pollution, and chemical fumes and dusts from the environment or workplace
- **Age.** Most people who have COPD are at least 40 years old when their symptoms begin.
- **Genetics.** This includes alpha-1 antitrypsin deficiency, which is a genetic condition. Also, smokers who get COPD are more likely to get it if they have a family history of COPD.
- **Asthma.** People who have [asthma](https://medlineplus.gov/asthma.html) [https://medlineplus.gov/asthma.html] have more risk of developing COPD than people who don't have asthma. But most people with asthma will **not** get COPD.

What are the symptoms of COPD?

At first, you may have no symptoms or only mild symptoms. As the disease gets worse, your symptoms usually become more severe. They can include:

- Frequent coughing [https://medlineplus.gov/cough.html] or a cough that produces a lot of mucus
- Wheezing or a whistling or squeaky sound when you breathe
- Shortness of breath, especially with physical activity
- Tightness in your chest
- Fatigue [https://medlineplus.gov/fatigue.html]

Some people with COPD get frequent respiratory infections such as colds

[https://medlineplus.gov/commoncold.html] and the flu [https://medlineplus.gov/flu.html] . In severe cases, COPD can cause weight loss, weakness in your lower muscles, and swelling [https://medlineplus.gov/edema.html] in your ankles, feet, and legs.

How is COPD diagnosed?

To find out if you have COPD, your health care provider may:

- Ask about your medical history, which includes asking about your symptoms
- Ask about your family history
- Ask if you have ever smoked or been exposed to secondhand smoke and other lung irritants
- Order various tests, such as lung function tests [https://medlineplus.gov/lab-tests/lung-function-tests/] , a chest x-ray [https://medlineplus.gov/xrays.html] or CT scan [https://medlineplus.gov/ctscans.html]
- Order blood tests, which may include an arterial blood gas test [https://medlineplus.gov/lab-tests/arterial-blood-gas-abg-test/] or an alpha-1 antitrypsin test [https://medlineplus.gov/lab-tests/alpha-1-antitrypsin-testing/]

What are the treatments for COPD?

There is no cure for COPD. However, treatments can help with symptoms, slow the progress of the disease, and improve your ability to stay active. There are also treatments to prevent or treat complications of the disease.

Treatments include:

- **Lifestyle changes**, such as:
 - Quitting smoking [https://medlineplus.gov/quittingsmoking.html] if you are a smoker. This is the most important step you can take to treat COPD.

- Avoiding secondhand smoke and places where you might breathe in other lung irritants.
- Following an eating plan that meets your nutritional needs.
- Increasing your physical activity as your condition allows. Ask your provider about how much physical activity you can do. Physical activity can strengthen the muscles that help you breathe and improve your overall wellness.
- **Medicines**, such as:
 - **Bronchodilators**, which relax the muscles around your airways. This helps open your airways and makes breathing easier. Most bronchodilators are taken through an inhaler. In more severe cases, the inhaler may also contain **steroids** [<https://medlineplus.gov/steroids.html>] to reduce inflammation.
 - **Vaccines** for the flu [<https://medlineplus.gov/flushot.html>] and pneumococcal pneumonia, since people with COPD are at higher risk for serious problems from these diseases
 - **Antibiotics** [<https://medlineplus.gov/antibiotics.html>] if you get a bacterial lung infection
 - **Biologics**, which are a type of medicine made from living sources, such as cells, tissues, and proteins. A biologic medicine may reduce inflammation in certain types of COPD
- **Oxygen therapy** [<https://medlineplus.gov/oxygentherapy.html>] , if you have severe COPD and low levels of oxygen in your blood. Oxygen therapy can help you breathe better. You may need extra oxygen all the time or only at certain times.
- **Pulmonary rehabilitation** [<https://medlineplus.gov/pulmonaryrehabilitation.html>] , which is a program that helps improve the well-being of people who have chronic breathing problems. It may include:
 - An exercise program
 - Disease management training
 - Nutritional counseling
 - Psychological counseling
- **Surgery**, usually as a last resort for people who have severe symptoms that have not gotten better with medicines:
 - For COPD that is mainly related to emphysema, there are surgeries that:
 - Remove damaged lung tissue.
 - Remove large air spaces (bullae) that can form when air sacs are destroyed. The bullae can interfere with breathing.
 - For severe COPD, some people may need a lung transplant.
[<https://medlineplus.gov/lungtransplantation.html>]

If you have COPD, it's important to know when and where to get help for your symptoms. You should get emergency care if you have severe symptoms, such as trouble catching your breath or talking. Call your provider if your symptoms are getting worse or if you have signs of an infection, such as a **fever** [<https://medlineplus.gov/fever.html>] .

Can COPD be prevented?

Since smoking causes most cases of COPD, the best way to prevent it is not to smoke. It's also important to try to avoid lung irritants such as secondhand smoke, air pollution, chemical fumes, and dusts.

NIH: National Heart, Lung, and Blood Institute

The information on this site should not be used as a substitute for professional medical care or advice. Contact a health care provider if you have questions about your health.

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